Question 1

A Generative AI Engineer must create and query a vector search index for a knowledge management system. What steps are essential to ensure accurate results?

1. Preprocess the documents, create embeddings for the text, store the embeddings in a vector store, and use nearest-neighbor search for queries.

Explanation

Correct: Preprocessing ensures clean data, embeddings enable semantic understanding, and nearest-neighbor search retrieves accurate results.

2. Use only keyword-based matching for queries.

Explanation

Incorrect: Keyword-based matching lacks the semantic depth provided by vector search.

3. Rely on raw document storage without embeddings.

Explanation

Incorrect: Raw document storage cannot support efficient vector search or semantic matching.

4. Skip document preprocessing and directly create embeddings.

Explanation

Incorrect: Preprocessing removes noise, improving the quality of embeddings and retrieval.

Overall explanation

A pipeline that preprocesses documents, creates embeddings, and queries using vector search ensures accurate and efficient results.

Question 2

A Generative AI Engineer at an electronics company just deployed a RAG application for customers to ask questions about products. However, they received feedback that the RAG response often returns information about an irrelevant product. What can the engineer do to improve the relevance of the RAG's response?

1. Use a different semantic similarity search algorithm.

Explanation

Incorrect: While this could help, assessing the retrieved context is the first step in diagnosing and improving relevance.

2. Use a different LLM to improve the generated response.

Explanation

Incorrect: The issue lies in retrieval, not the LLM's generative capabilities.

3. Implement caching for frequently asked questions.

Explanation

Incorrect: While caching improves speed, it does not address retrieval quality or relevance issues.

4. Assess the quality of the retrieved context.

Explanation

Correct: Checking the quality of retrieved documents ensures only relevant information is fed into the LLM, improving response accuracy.

Overall explanation

Evaluating the quality of retrieved context ensures that only accurate and relevant information is passed to the LLM, solving relevance issues effectively.

Question 3

A Generative AI Engineer is tasked with writing chunked user feedback data into Delta Lake tables. Which operation sequence ensures optimal data preparation and storage?

1. Chunk the text first → Write the chunks into Delta Lake → Define the schema later.

Explanation

Incorrect: Schema must be defined before writing to ensure proper organization.

2. Skip chunking and directly write the feedback into Delta Lake.

Explanation

Incorrect: Unstructured data without chunking limits downstream processing efficiency.

3. Write all feedback data as one large chunk into Delta Lake.

Explanation

Incorrect: Storing data as one large chunk reduces query performance and accessibility.

4. Define schema for feedback data → Chunk the text → Write chunks into Delta Lake tables.

Explanation

Correct: Defining the schema first ensures structured data storage, improving accessibility and consistency.

Overall explanation

Defining the schema first ensures data consistency, and chunking facilitates efficient storage and retrieval in Delta Lake tables.

Question 4

A Generative AI Engineer needs to evaluate the output of a chatbot that generates legal advice. The outputs often include ambiguous statements. What method should they use to identify and resolve these issues?

1. Use a summarization model to shorten responses.

Explanation

Incorrect: Summarization may remove necessary context, increasing ambiguity.

2. Measure BLEU scores for the chatbot's responses.

Explanation

Incorrect: BLEU evaluates text similarity but does not address ambiguity or accuracy.

3. Train the chatbot on more extensive datasets.

Explanation

Incorrect: Larger datasets may improve general performance but do not specifically address ambiguity.

4. Qualitatively assess the responses for clarity and accuracy, then refine prompts to reduce ambiguity.

Explanation

Correct: Qualitative assessment ensures outputs align with legal standards, and prompt refinement addresses identified issues.

Overall explanation

Qualitative assessments and prompt refinement directly tackle the root causes of ambiguous responses, ensuring improved outputs.

Question 5

A Generative AI Engineer is tasked with building a prompt for a chatbot that answers customer product questions. The chatbot should prioritize factual accuracy over creativity. What should the prompt include?

1. No instructions, relying on the model's default behavior.

Explanation

Incorrect: Default behavior may not align with the requirement for factual accuracy.

2. Instructions to "Be creative in your response."

Explanation

Incorrect: Encouraging creativity could compromise factual accuracy.

3. General guidance like "Answer the query."

Explanation

Incorrect: General prompts lack specificity and may lead to inconsistent or inaccurate responses.

4. Explicit instructions such as “Provide concise, factual answers to customer questions.”

Explanation

Correct: Clear instructions help the model focus on generating fact-based responses over creative or imaginative ones.

Overall explanation

Clear and explicit prompts guide the LLM to produce accurate, consistent responses tailored to the application's needs.

Question 6

A Generative AI Engineer is selecting a model for generating secure medical chatbot responses. Which feature is critical for ensuring data privacy?

1. The model should allow unlimited external API calls.

Explanation

Incorrect: Unlimited API calls increase the risk of data breaches.

2. The model should prioritize response diversity.

Explanation

Incorrect: Response diversity does not directly address data privacy.

3. The model should always prioritize speed over accuracy.

Explanation

Incorrect: Speed does not ensure compliance with privacy standards.

4. The model should not store user input or generate responses based on external data storage.

Explanation

Correct: Ensuring the model does not store input prevents sensitive data leaks and ensures compliance with privacy standards.

Overall explanation

A model that avoids storing user input and external dependencies ensures compliance with privacy standards in sensitive domains like healthcare.

Question 7

A Generative AI Engineer is debugging a system where an LLM provides incomplete responses to user queries. What is the most likely cause?

1. The model is too large for the task.

Explanation

Incorrect: Model size does not directly affect response completeness.

2. The temperature setting is too high.

Explanation

Incorrect: Higher temperature affects randomness but not completeness.

3. The prompt does not provide clear instructions or adequate context.

Explanation

Correct: Incomplete or unclear prompts often lead to incomplete responses from LLMs.

4. The user queries are too short.

Explanation

Incorrect: Query length alone does not guarantee response quality.

Overall explanation

Ensuring prompts are detailed and clear helps guide the LLM to generate complete and accurate responses.

Question 8

A Generative AI Engineer must create a pipeline that retrieves financial reports and summarizes key performance metrics. The system must handle thousands of reports efficiently. What is the correct sequence for designing this pipeline?

1. 1.Fine-tune a summarization model → 2. Generate embeddings → 3. Build a retriever → 4. Deploy to production.

Explanation

Incorrect: Summarization models are applied after retrieval, not before embedding generation.

2. 1.Preprocess all reports → 2. Deploy a retriever → 3. Fine-tune a language model → 4. Summarize outputs.

Explanation

Incorrect: Preprocessing and retrieval must include indexing for effective querying.

3. 1.Deploy the retriever → 2. Store raw reports in a database → 3. Summarize retrieved reports → 4. Index the database.

Explanation

Incorrect: Indexing must occur before retrieval for efficient querying.

4. 1.Create embeddings for all financial reports → 2. Store embeddings in a Vector Search index → 3. Build a retriever → 4. Deploy a summarization model for retrieval outputs.

Explanation

Correct: This sequence ensures that documents are indexed for retrieval before being summarized, optimizing both retrieval and generation processes.

Overall explanation

Embedding generation, vector indexing, retrieval, and summarization ensure the pipeline is scalable and optimized for financial document processing.

Question 9

A Generative AI Engineer is designing a prompt for a QA system to answer questions about historical events. The response must include dates, key figures, and outcomes. How should the prompt be structured?

1. Explain the causes of the historical event.

Explanation

Incorrect: Causes alone do not fulfill the requirement for detailed answers.

2. Provide a summary of the historical event.

Explanation

Incorrect: Summaries may skip critical details like dates or key figures.

3. Provide a general overview of historical events.

Explanation

Incorrect: Overviews are too broad and lack specificity.

4. Answer the question with details about dates, key figures, and outcomes.

Explanation

Correct: A clear, specific prompt ensures the inclusion of all required details.

Overall explanation

A clear prompt specifying required fields ensures the QA system generates accurate and comprehensive responses.

Question 10

Which metric is most suitable for evaluating the accuracy of summaries generated by an LLM?

1. ROUGE score

Explanation

Correct: ROUGE compares the generated summary with a reference summary to measure overlap and relevance.

2. F1 score

Explanation

Incorrect: F1 is used in classification tasks and is not directly applicable to summaries.

3. BLEU score

Explanation

Incorrect: BLEU is primarily used for translation tasks, not summarization.

4. Perplexity

Explanation

Incorrect: Perplexity measures language model confidence but not summary quality.

Overall explanation

The ROUGE score is specifically designed to evaluate summarization tasks, making it the best metric for assessing summary accuracy.

Question 11

A Generative AI Engineer must register a custom LLM to Unity Catalog using MLflow. What are the key steps in this process?

1. Register the model directly in Unity Catalog without logging it in MLflow.

Explanation

Incorrect: MLflow provides critical logging and metadata management capabilities required before registration.

2. Use only the MLflow tracking server without registering the model.

Explanation

Incorrect: Registration to Unity Catalog is required for centralized governance and discovery.

3. Log the model in MLflow, configure its metadata (e.g., schema and description), and register it to Unity Catalog for governance and discovery.

Explanation

Correct: Logging and metadata configuration ensure the model is properly tracked and registered for central access and governance.

4. Skip metadata configuration during registration.

Explanation

Incorrect: Metadata is essential for model discoverability and usage tracking.

Overall explanation

Proper logging, metadata configuration, and registration ensure that the model is discoverable and governed effectively within Unity Catalog.

Question 12

A Generative AI Engineer is tasked with building a Vector Search index to handle document queries for a knowledge base. What steps are required to create and query this index?

1. Skip preprocessing and store raw document text in a vector store for querying.

Explanation

Incorrect: Without preprocessing, documents may include noise, degrading retrieval quality.

2. Use a retrieval model directly on raw documents without embeddings.

Explanation

Incorrect: Retrievers require embeddings to semantically represent document content for effective search.

3. Use embeddings to encode queries but not documents.

Explanation

Incorrect: Both documents and queries must be embedded for semantic alignment during search.

4. Preprocess documents, embed them using an embedding model, store the embeddings in a vector store, and use a retriever for queries.

Explanation

Correct: Preprocessing, embedding, and retrieval are essential for building and querying an effective Vector Search index.

Overall explanation

Creating a Vector Search index requires preprocessing, embedding, storing in a vector store, and using retrievers for query handling.

Question 13

A Generative AI Engineer must assess the scalability of a production LLM application in handling increasing query volumes. What metrics are most critical?

1. Token usage and cost per query.

Explanation

Incorrect: While important, these metrics do not measure scalability directly.

2. Model size and parameter count.

Explanation

Incorrect: Model size alone does not evaluate scalability.

3. Throughput and latency under varying workloads.

Explanation

Correct: These metrics directly assess the system's ability to handle increasing query volumes efficiently.

4. BLEU and perplexity.

Explanation

Incorrect: These metrics evaluate language quality, not system scalability.

Overall explanation

Throughput and latency are critical for evaluating how well a production LLM application scales under increasing workloads.

Question 14

A Generative AI Engineer is evaluating an embedding model for a system that recommends job candidates based on resumes and job descriptions. The model has a 512-token context length. How should they adapt the system for longer resumes?

1. Use a summarization model to reduce resume length.

Explanation

Incorrect: Summarization risks omitting critical details.

2. Chunk resumes into smaller sections within the 512-token limit and process each chunk independently.

Explanation

Correct: Chunking ensures compliance with token limits while preserving content relevance.

3. Ignore longer resumes and process only those within the token limit.

Explanation

Incorrect: Ignoring resumes reduces the system's effectiveness and fairness.

4. Train a new embedding model with a longer context length.

Explanation

Incorrect: Training new models is resource-intensive and not always feasible.

Overall explanation

Chunking resumes into smaller sections allows the system to process longer inputs while adhering to the token limit.

Question 15

A Generative AI Engineer is using inference logging to monitor a RAG application. What insights can inference logging provide?

1. BLEU scores and perplexity for language outputs.

Explanation

Incorrect: BLEU and perplexity assess language quality but are not relevant to inference logging.

2. Model training loss over time.

Explanation

Incorrect: Training loss is a metric for model development, not deployment monitoring.

3. Token usage and model size.

Explanation

Incorrect: Token usage and model size are unrelated to inference logging insights.

4. Query patterns, response times, and error occurrences.

Explanation

Correct: Inference logging tracks query behavior, latency, and errors, offering insights into application performance and user interactions.

Overall explanation

Inference logging provides actionable data about query patterns, latency, and errors, helping to optimize application performance.

Question 16

A Generative AI Engineer is designing an AI assistant for lawyers. The assistant must retrieve legal documents and generate concise summaries. What is the most important consideration for ensuring accurate retrieval?

1. Build a rule-based filtering system.

Explanation

Incorrect: Rule-based systems are inflexible and unsuitable for nuanced legal queries.

2. Fine-tune the LLM on legal datasets.

Explanation

Incorrect: Fine-tuning improves summarization but does not address document retrieval.

3. Use a keyword-based search algorithm.

Explanation

Incorrect: Keyword-based retrieval may miss documents that are contextually relevant but lack exact keyword matches.

4. Implement a vector database for semantic search.

Explanation

Correct: Vector databases enable semantic retrieval, ensuring the system retrieves documents based on meaning rather than exact keywords.

Overall explanation

Semantic search via a vector database ensures the system retrieves relevant legal documents, providing accurate inputs for generating concise summaries.

Question 17

A Generative AI Engineer must design a prompt for a chatbot that provides software installation instructions. The response should include step-by-step instructions and warnings for common issues. What should the prompt include?

1. Include only the main steps of the installation process.

Explanation

Incorrect: Focusing only on main steps may omit important warnings or troubleshooting.

2. Provide detailed installation instructions, including common warnings and troubleshooting tips.

Explanation

Correct: Explicitly asking for detailed steps and warnings ensures the response is comprehensive.

3. Summarize the installation process for the software.

Explanation

Incorrect: Summaries may skip critical steps or details.

4. Provide a concise explanation of software features.

Explanation

Incorrect: Features are unrelated to installation instructions.

Overall explanation

A clear, detailed prompt ensures the chatbot provides step-by-step instructions with warnings, aligning with user expectations.

Question 18

A Generative AI Engineer needs to extract customer feedback from PDFs containing tables, images, and text. What tool is most appropriate for this task?

1. pdfplumber for extracting structured text and tables from PDFs.

Explanation

Correct: pdfplumber is optimized for extracting structured text and tables, making it ideal for processing mixed-content PDFs.

2. PyPDF2 for reading and extracting PDF content.

Explanation

Incorrect: PyPDF2 cannot handle tables or images effectively.

3. BeautifulSoup for parsing PDF content.

Explanation

Incorrect: BeautifulSoup is designed for HTML, not PDF parsing.

4. pytesseract for OCR extraction.

Explanation

Incorrect: pytesseract is used for scanned images, not text in PDFs.

Overall explanation

pdfplumber is the best tool for extracting structured content like text and tables from PDFs, ensuring accurate data retrieval.

Question 19

A Generative AI Engineer is tasked with selecting tools for a pipeline that converts text-based user reviews into categorized insights. The categories include product quality, delivery, and customer service. What components are critical?

1. A summarization model to condense user reviews.

Explanation

Incorrect: Summarization does not produce specific categorizations.

2. An embedding model to represent reviews semantically.

Explanation

Incorrect: While embeddings capture meaning, they do not categorize reviews directly.

3. A retrieval system to fetch similar reviews.

Explanation

Incorrect: Retrieval does not classify or generate categorized insights.

4. A classification model to categorize user reviews based on predefined categories.

Explanation

Correct: Classification models assign reviews to specific categories effectively.

Overall explanation

A classification model ensures reviews are assigned to the correct categories, enabling actionable insights from user feedback.

Question 20

A Generative AI Engineer is deploying a chatbot for a healthcare application. How can the engineer ensure the chatbot does not generate private patient information in its outputs?

1. Encrypt all chatbot responses before returning them to the user.

Explanation

Incorrect: Encryption secures responses during transmission but does not prevent private data inclusion.

2. Use input filtering and response validation to block private data during processing and output generation.

Explanation

Correct: Filtering inputs and validating outputs ensures that private data is not included in the chatbot's responses, protecting user privacy.

3. Train the chatbot on anonymized datasets only.

Explanation

Incorrect: While helpful, training on anonymized data does not address private data inclusion in inputs.

4. Limit the chatbot's response length to prevent detailed outputs.

Explanation

Incorrect: Limiting response length does not specifically target private data issues.

Overall explanation

Input filtering and response validation ensure that private information is excluded from chatbot outputs, complying with privacy requirements.

Question 21

A Generative AI Engineer needs to develop a pipeline that identifies and extracts customer intents from chatbot queries. Which embedding model should they select?

1. Word2Vec

Explanation

Incorrect: Word2Vec creates embeddings for individual words but does not capture full sentence meaning effectively.

2. BERT-base for classification

Explanation

Incorrect: Classification models like BERT-base are not optimized for generating embeddings for semantic search.

3. Sentence Transformers

Explanation

Correct: Sentence Transformers are optimized for capturing semantic relationships in text, making them suitable for identifying user intent.

4. GloVe

Explanation

Incorrect: GloVe embeddings are pre-trained on word-level data and are less suitable for sentence-level tasks.

Overall explanation

Sentence Transformers excel in generating embeddings for entire sentences, making them ideal for tasks like intent detection in chatbot queries.

Question 22

A Generative AI Engineer is optimizing document retrieval for a customer support chatbot. User feedback indicates that irrelevant results frequently appear. What should the engineer focus on improving?

1. Document length filtering

Explanation

Document length filtering may help in optimizing document retrieval by filtering out excessively long or short documents. However, the user feedback indicates that irrelevant results frequently appear, which suggests that the issue lies more with the relevance of the results rather than the length of the documents. Therefore, improving document length filtering may not directly address the problem of irrelevant results.

2. Response generation diversity

Explanation

Response generation diversity is related to the variety of responses generated by the chatbot. While diversity in responses is important for engaging users and providing a range of answers, the primary issue mentioned in the user feedback is the presence of irrelevant results. Therefore, focusing on response generation diversity may not directly address the problem of irrelevant search results in this context.

3. Recall

Explanation

Recall is the measure of the completeness of the retrieved results. While recall is important in ensuring that all relevant results are retrieved, the user feedback specifically mentions irrelevant results appearing frequently. Therefore, the engineer should focus on improving precision rather than recall in this scenario.

4. Precision

Explanation

Precision is the measure of the accuracy of the retrieved results. In this case, the engineer should focus on improving precision to ensure that the search results are relevant to the user's query. By increasing precision, the engineer can reduce the number of irrelevant results appearing in the chatbot.

Overall explanation

Focusing on precision ensures the chatbot retrieves only relevant content, improving user satisfaction and system accuracy.

Question 23

A Generative AI Engineer must create a prompt that ensures consistent responses from an LLM when summarizing financial reports. What is the most effective prompt strategy?

1. Provide clear instructions and examples in the prompt, such as “Summarize the key financial figures and trends clearly and concisely.”

Explanation

Correct: Including specific instructions and examples helps guide the model, ensuring consistent and high-quality responses.

2. Use a short, vague prompt like “Summarize the report.”

Explanation

Incorrect: Vague prompts lack clarity and can lead to inconsistent or irrelevant outputs.

3. Use multiple prompts with contradictory instructions.

Explanation

Incorrect: Contradictory instructions confuse the model and reduce response quality.

4. Allow the model to interpret the task without guidance.

Explanation

Incorrect: Without guidance, the LLM may produce incomplete or irrelevant summaries.

Overall explanation

A well-structured prompt with explicit instructions and examples improves response consistency and aligns outputs with expectations.

Question 24

A Generative AI Engineer is tasked with extracting text from HTML files containing product catalogs. What is the most suitable Python package for this task? **(Choose two)**

1. BeautifulSoup for parsing HTML content and extracting text.

Explanation

Correct: BeautifulSoup is designed for HTML parsing and text extraction.

2. pytesseract for OCR tasks.

Explanation

Incorrect: OCR tools like pytesseract are unnecessary for HTML files.

3. pdfplumber for text extraction.

Explanation

Incorrect: pdfplumber is not relevant for HTML parsing tasks.

4. PyPDF2 for extracting text from PDFs.

Explanation

Incorrect: PyPDF2 is not suitable for parsing HTML files.

5. LXML for faster and efficient parsing of large HTML files.

Explanation

Correct: LXML is optimized for parsing large files, making it ideal for high-volume tasks.

Overall explanation

Combining BeautifulSoup and LXML ensures robust text extraction from HTML files, meeting the task requirements.

Question 25

A Generative AI Engineer is designing a pipeline for multi-turn dialogue in a customer support chatbot. The chatbot often forgets earlier turns. What should they implement?

1. Reset the context after each turn.

Explanation

Incorrect: Resetting context results in disjointed and unhelpful responses.

2. Increase the token limit.

Explanation

Incorrect: Token limit affects input size but does not resolve multi-turn context issues.

3. State tracking to retain the context of previous dialogue turns.

Explanation

Correct: State tracking helps maintain coherence by preserving the context of the conversation.

4. Use a lower temperature setting.

Explanation

Incorrect: Lowering temperature affects randomness but not context retention.

Overall explanation

State tracking ensures the chatbot remembers previous conversation turns, improving coherence and user experience.

Question 26

A Generative AI Engineer is tasked with creating a simple chain in LangChain that processes user queries about product specifications. The chain must return precise answers from a structured product database. What components are required? **(Choose two)**

1. A rule-based filtering system for product categories.

Explanation

Incorrect: Rule-based filters are rigid and limit the system's flexibility.

2. A summarization model to shorten user queries.

Explanation

Incorrect: Summarization is unnecessary for retrieving specific product specifications.

3. A pre-trained classification model for query analysis.

Explanation

Incorrect: Classification models are not required for this task.

4. A structured prompt template to format user queries.

Explanation

Correct: Structured prompts guide the chain to generate precise responses based on the database.

5. A database retriever to fetch relevant product specifications.

Explanation

Correct: A retriever ensures dynamic access to relevant information from the product database.

Overall explanation

Combining retrievers and structured prompts ensures accurate and precise responses for product-related queries.

Question 27

A Generative AI Engineer is tasked with reviewing the licensing requirements of a dataset containing medical records used in a generative application. What steps should they take to ensure legal compliance? **(Choose two)**

1. Verify that the dataset complies with HIPAA or relevant data privacy regulations.

Explanation

Correct: Ensuring compliance with regulations like HIPAA is critical for handling sensitive medical data.

2. Assume non-commercial use does not require licensing checks.

Explanation

Incorrect: Legal compliance is necessary regardless of the intended use case.

3. Check the licensing terms to confirm the dataset can be used for commercial purposes.

Explanation

Correct: Reviewing licensing terms prevents unauthorized use of the dataset.

4. Use the dataset as long as it is publicly available.

Explanation

Incorrect: Public availability does not equate to permissible use without verifying licensing and privacy compliance.

5. Ensure the LLM’s outputs include disclaimers about data origins.

Explanation

Incorrect: Disclaimers do not replace the need for licensing and privacy compliance.

Overall explanation

Verifying privacy regulations and licensing terms ensures the dataset is legally compliant and suitable for use in the application.

Question 28

A Generative AI Engineer must sequence the steps to deploy a RAG application endpoint. What is the correct order? **(Choose two)**

1. Deploy the endpoint first and configure embeddings later.

Explanation

Incorrect: Embeddings and vector stores must be prepared before deployment.

2. Use hardcoded responses instead of a retriever.

Explanation

Incorrect: Hardcoded responses lack flexibility for dynamic queries.

3. Skip preprocessing and deploy documents directly to the endpoint.

Explanation

Incorrect: Preprocessing removes noise and ensures clean data for retrieval.

4. Preprocess documents and create a vector store before deploying the endpoint.

Explanation

Correct: Preprocessing and vector store creation ensure clean and retrievable data for endpoint deployment.

5. Rely only on the LLM without embedding or vector search.

Explanation

Incorrect: Embeddings and vector search are critical for RAG applications.

6. Train the embedding model, preprocess documents, create a vector store, and deploy the endpoint.

Explanation

Correct: This sequence ensures data is prepared, stored, and served through an endpoint effectively.

Overall explanation

Preprocessing, embeddings, and vector store creation are essential steps for deploying an effective RAG application endpoint.

Question 29

A Generative AI Engineer needs to register a machine learning model in Unity Catalog for deployment. What benefits does this provide?

1. Centralized model governance, versioning, and access control.

Explanation

Correct: Unity Catalog integrates with MLflow to provide features like tracking, lineage, and governance for machine learning models.

2. Reduced model inference time.

Explanation

Incorrect: Unity Catalog manages governance and registry but does not impact inference speed.

3. Automated hyperparameter tuning.

Explanation

Incorrect: Unity Catalog tracks model metadata but does not handle tuning.

4. Fine-tuning capabilities for custom tasks.

Explanation

Incorrect: Fine-tuning is a separate process not handled by Unity Catalog.

Overall explanation

Unity Catalog centralizes model governance, enabling tracking, versioning, and access control for robust machine learning workflows.

Question 30

A Generative AI Engineer is evaluating the retrieval performance of a RAG system. Which metrics are most relevant to this evaluation?

1. Latency to measure response time.

Explanation

Incorrect: Latency measures speed but does not assess relevance or accuracy of retrieval.

2. BLEU scores for retrieval evaluation.

Explanation

Incorrect: BLEU evaluates text similarity in translation or summarization tasks, not retrieval.

3. Perplexity to measure response fluency.

Explanation

Incorrect: Perplexity evaluates language generation, not retrieval performance.

4. Precision and recall to measure relevance and coverage of retrieved results.

Explanation

Correct: Precision evaluates the relevance of retrieved documents, while recall assesses how many relevant documents are retrieved.

Overall explanation

Precision and recall are key metrics for evaluating the relevance and coverage of retrieved results in RAG systems.

Question 31

A Generative AI Engineer is evaluating the performance of a RAG application that retrieves documents and generates summaries. The retrieval step has high recall but low precision. What does this indicate, and how should it be addressed?

1. The system has low latency, which reduces retrieval accuracy.

Explanation

Incorrect: Latency impacts speed, not the balance between precision and recall.

2. The system retrieves all relevant documents but excludes others, requiring larger batch sizes.

Explanation

Incorrect: Recall measures the proportion of relevant documents retrieved, not batch size.

3. The summarization step is producing incomplete summaries.

Explanation

Incorrect: Summarization quality is unrelated to retrieval metrics like precision and recall.

4. The system retrieves many relevant documents but also includes irrelevant ones, requiring better filtering or embeddings.

Explanation

Correct: High recall with low precision indicates the need to refine retrieval to reduce irrelevant results, improving overall relevance.

Overall explanation

High recall but low precision means the system retrieves many irrelevant documents, which can be addressed with better embeddings or retrieval filters.

Question 32

A Generative AI Engineer is tasked with building a Retrieval-Augmented Generation (RAG) system for a healthcare organization. The system needs to retrieve patient-specific documents and summarize key findings securely. What are the critical components for building this pipeline? **(Choose four)**

1. Low-latency GPU infrastructure for real-time queries.

Explanation

Correct: GPUs are essential for efficiently handling the computational demands of retrieval and summarization in real-time applications.

2. Embedding model trained on healthcare datasets.

Explanation

Correct: An embedding model trained on domain-specific data ensures that document embeddings align with medical terminology and user queries.

3. Secure document retriever with access controls.

Explanation

Correct: A secure retriever prevents unauthorized access to sensitive patient data and ensures compliance with privacy regulations like HIPAA.

4. Summarizer model optimized for clinical reports.

Explanation

Correct: Summarization models tailored for clinical text ensure concise and relevant outputs for healthcare professionals.

5. Classification model for tagging medical conditions.

Explanation

Incorrect: While classification may help identify conditions, it is not a mandatory component for RAG pipelines.

Overall explanation

RAG systems for healthcare require domain-specific embedding models, secure retrievers, summarization capabilities, and high-performance compute resources to ensure accuracy, compliance, and efficiency.

Question 33

A Generative AI Engineer is tasked with registering an LLM to Unity Catalog using MLflow. What information must be configured during registration?

1. Use only MLflow without registering the model to Unity Catalog.

Explanation

Incorrect: Unity Catalog provides centralized management and governance for models.

2. Model metadata such as name, version, schema, and tags for tracking and governance.

Explanation

Correct: Configuring metadata ensures the model is properly registered and discoverable in Unity Catalog.

3. Only the model name and version without schema or tags.

Explanation

Incorrect: Schema and tags are critical for tracking and usability.

4. Skip metadata configuration and register the model with default settings.

Explanation

Incorrect: Default settings lack sufficient information for governance and tracking.

Overall explanation

Metadata like schema and tags ensures models are registered with all necessary details for effective governance and tracking.

Question 34

A Generative AI Engineer is developing a customer support chatbot for an e-commerce platform. The chatbot must provide answers to common queries while escalating unresolved issues to human agents. What should the engineer prioritize in the design?

1. Focus on embedding models to understand user queries semantically.

Explanation

Incorrect: While embeddings aid query understanding, they do not address unresolved query escalation.

2. Train the chatbot on a large dataset of historical conversations.

Explanation

Incorrect: Training improves general understanding but does not implement escalation mechanisms.

3. Use rule-based logic for predefined query handling.

Explanation

Incorrect: Rule-based systems lack flexibility for dynamic customer queries.

4. Implement a fallback mechanism to escalate unresolved queries to human agents.

Explanation

Correct: Fallback mechanisms ensure seamless escalation when the chatbot cannot handle a query.

Overall explanation

Fallback mechanisms enhance user experience by ensuring unresolved queries are appropriately escalated to human agents.

Question 35

A Generative AI Engineer is building a chatbot for a travel agency. The chatbot must recommend travel destinations based on weather preferences, budget, and activities. What is the first step in pipeline design?

1. Select an LLM trained on travel data for recommendations.

Explanation

Incorrect: Model selection is secondary to defining input-output relationships.

2. Create a database of destinations and activities.

Explanation

Incorrect: Databases support recommendations but require predefined pipeline structure.

3. Gather feedback on early recommendations.

Explanation

Incorrect: Feedback is important later for optimization but not during initial pipeline design.

4. Define input fields like budget, preferred_weather, and activities and output fields like destination and activity_recommendations. Example: Input: { "budget": "$3000", "preferred_weather": "tropical", "activities": "adventure" } → Output: { "destination": "Thailand", "activity_recommendations": "snorkeling, jungle trekking" }.

Explanation

Correct: Clearly defining inputs and outputs ensures the pipeline aligns with business goals and user requirements.

Overall explanation

Clearly defining inputs like budget and outputs like destination recommendations ensures alignment with user requirements and guides the pipeline's architecture.

Question 36

A Generative AI Engineer must preprocess a dataset of forum posts for a support chatbot. Many posts include hyperlinks and embedded media. What preprocessing step should the engineer take?

1. Retain hyperlinks to preserve additional context.

Explanation

Incorrect: Hyperlinks are typically irrelevant and may confuse the model.

2. Summarize posts without removing hyperlinks or media.

Explanation

Incorrect: Summarization without cleaning may retain noise, reducing dataset quality.

3. Exclude posts containing hyperlinks or media entirely.

Explanation

Incorrect: Excluding such posts risks losing valuable textual content alongside irrelevant media.

4. Remove hyperlinks and embedded media, retaining only the text content.

Explanation

Correct: Removing non-textual elements ensures the dataset focuses on meaningful language content, which the chatbot can process effectively.

Overall explanation

Removing hyperlinks and media ensures the dataset remains focused on language content, optimizing the chatbot's training and performance.

Question 37

A Generative AI Engineer is tasked with coding a LangChain-based RAG application that queries a large dataset of financial reports. The dataset includes structured and unstructured data. What components should the chain include? **(Choose two)**

1. A retriever to fetch relevant data from the vector store.

Explanation

Correct: Retrievers enable semantic search and retrieval of relevant data.

2. A classification model to categorize financial queries.

Explanation

Incorrect: Classification does not directly aid retrieval in this scenario.

3. A pre-trained sentiment analysis model.

Explanation

Incorrect: Sentiment analysis is irrelevant for querying financial reports.

4. A summarization model to condense retrieval results.

Explanation

Incorrect: Summarization is not necessary for retrieving precise information from the dataset.

5. An embedding model to process both structured and unstructured data.

Explanation

Correct: Embedding models ensure that all data types are semantically represented for retrieval.

Overall explanation

Retrievers and embedding models ensure accurate and efficient query handling in the financial reports dataset.

Question 38

When developing an LLM application, it's crucial to ensure that the data used for training the model complies with licensing requirements to avoid legal risks. Which action is NOT appropriate to avoid legal risks?

1. Use any available data you personally created which is completely original and you can decide what license to use.

Explanation

Incorrect: Original data created by you is legally safe as long as you control its licensing.

2. Only use data explicitly labeled with an open license and ensure the license terms are followed.

Explanation

Incorrect: Following license terms for labeled open data avoids legal risks.

3. Reach out to the data curators directly before you have started using the trained model to let them know.

Explanation

Correct: Contacting curators after starting to use their data is inappropriate and exposes you to legal risk. Always clarify terms before use.

4. Reach out to the data curators directly after you have started using the trained model to let them know.

Explanation

Correct: Contacting curators after using their data does not retroactively grant permissions and could lead to legal consequences.

Overall explanation

Always establish permissions or verify licensing terms before using third-party data in training to avoid legal risks.

Question 39

A Generative AI Engineer must evaluate an LLM for summarizing financial news articles. The system must generate accurate, concise, and timely summaries. What metrics should guide model evaluation and deployment? **(Choose Four)**

1. ROUGE for evaluating content relevance and overlap with reference summaries.

Explanation

Correct: ROUGE measures the relevance and quality of generated summaries.

2. Latency to ensure the system delivers summaries quickly.

Explanation

Correct: Low latency ensures timely delivery, crucial for time-sensitive financial news.

3. BLEU to compare word sequences in generated summaries.

Explanation

Correct: BLEU evaluates the alignment of generated summaries with reference outputs for accuracy.

4. Token usage per query to track computational cost.

Explanation

Incorrect: While cost-related, token usage does not directly evaluate accuracy or timeliness.

5. Perplexity to assess the fluency of summaries.

Explanation

Correct: Perplexity measures the naturalness and readability of the summaries, enhancing user experience.

Overall explanation

A combination of latency, ROUGE, BLEU, and perplexity ensures the system is fast, accurate, and user-friendly for financial news summarization.

Question 40

A Generative AI Engineer is building a generative application for processing user-provided content. The application must protect against malicious user inputs such as injection attacks. What guardrail technique should they use?

1. Use a summarization model to preprocess user inputs.

Explanation

Incorrect: Summarization is not a reliable method to mitigate injection risks.

2. Allow unrestricted user inputs to encourage flexibility.

Explanation

Incorrect: Unrestricted inputs expose the application to security vulnerabilities.

3. Sanitize user inputs by removing special characters and validating data formats.

Explanation

Correct: Sanitizing inputs prevents malicious payloads, such as SQL injection, from being processed by the application.

4. Log and monitor user inputs for suspicious patterns.

Explanation

Incorrect: Monitoring alone does not prevent malicious inputs from being executed.

Overall explanation

Input sanitization effectively neutralizes harmful payloads, ensuring application security against malicious inputs.

Question 41

A Generative AI Engineer must generate multiple-choice questions from a large corpus of educational material. The questions should assess conceptual understanding and application of knowledge. What approach should they take?

1. Use a summarization model to distill the material into key points.

Explanation

Incorrect: Summarization does not produce structured questions.

2. Train a classification model to generate answers from the material.

Explanation

Incorrect: Classification models are not designed to generate multiple-choice questions.

3. Use a prompt template that specifies question structure, options, and the correct answer.

Explanation

Correct: Prompt templates guide the LLM to generate well-structured and accurate multiple-choice questions.

4. Fine-tune the LLM on educational datasets.

Explanation

Incorrect: Fine-tuning aids domain knowledge but does not ensure multiple-choice question generation.

Overall explanation

Prompt templates ensure that the LLM generates questions aligned with the educational goals, including structured options and correct answers.

Question 42

A Generative AI Engineer is designing a customer service chatbot to handle product return requests. The chatbot must ensure that responses are both accurate and polite. What approach should they use to implement this?

1. Fine-tune the LLM on a polite tone dataset and return policy information.

Explanation

Correct: Fine-tuning ensures both tone and domain-specific knowledge align with application needs.

2. Add politeness rules into the system as post-processing filters.

Explanation

Incorrect: Filters can flag issues but do not inherently improve response quality or accuracy.

3. Rely on summarization techniques to handle user inputs.

Explanation

Incorrect: Summarization does not improve politeness or ensure accuracy for return requests.

4. Use a generic pre-trained model for handling responses.

Explanation

Incorrect: Generic models lack domain knowledge and may not handle tone requirements.

Overall explanation

Fine-tuning aligns the LLM's tone and knowledge base, ensuring polite and accurate responses for customer service scenarios.

Question 43

A Generative AI Engineer is working with a dataset containing both primary and secondary healthcare documents for a medical chatbot. What type of documents should they prioritize for ensuring factual accuracy?

1. Summaries of medical textbooks.

Explanation

Incorrect: Summaries are secondary sources and may not provide the most up-to-date or comprehensive information.

2. Peer-reviewed medical research papers and government health guidelines.

Explanation

Correct: Peer-reviewed research papers and government guidelines are credible, ensuring accurate and trustworthy chatbot responses.

3. Health-related social media posts.

Explanation

Incorrect: Social media content is subjective and lacks authoritative validation.

4. Patient blogs and forum discussions.

Explanation

Incorrect: Blogs and forums often contain unverified information and personal anecdotes unsuitable for medical applications.

Overall explanation

Using credible sources like peer-reviewed papers and government guidelines ensures reliable and factual medical chatbot responses.

Question 44

To reduce hallucinations in LLM responses, what is the most effective approach when implementing a retrieval-augmented generation (RAG) pipeline?

1. Use a high-quality document retrieval system to provide accurate context for the model.

Explanation

Correct: High-quality document retrieval anchors responses in factual data, minimizing hallucinations.

2. Train the model on a larger dataset.

Explanation

Incorrect: A larger training dataset does not guarantee reduced hallucinations during inference.

3. Shorten the length of model responses.

Explanation

Incorrect: Short responses do not inherently improve factual accuracy.

4. Increase the temperature parameter during inference.

Explanation

Incorrect: Increasing temperature adds randomness and can lead to more hallucinations.

Overall explanation

Grounding model responses in high-quality, retrieved context ensures factual outputs, which is critical for reducing hallucinations.

Question 45

A Generative AI Engineer must build a system to generate educational content summaries for students. The summaries should include three fields: key concepts, examples, and practical applications. What is the best pipeline design?

1. Use a summarization model to condense content into shorter text.

Explanation

Incorrect: Summarization does not generate structured outputs with specific fields.

2. Use a rule-based system to extract predefined educational elements.

Explanation

Incorrect: Rule-based systems lack flexibility and adaptability for diverse content.

3. Use an LLM to generate summaries dynamically, structured into key concepts, examples, and applications.

Explanation

Correct: LLMs dynamically generate structured, educational content summaries tailored to user needs.

4. Use an embedding model to represent educational text semantically.

Explanation

Incorrect: Embeddings alone do not create structured summaries.

Overall explanation

LLMs dynamically generate structured educational content, aligning with requirements for fields like key concepts and applications.